ONLINE MASTERS IN **DATA SCIENCE**

DSC 257R - UNSUPERVISED LEARNING

THE BINOMIAL DISTRIBUTION

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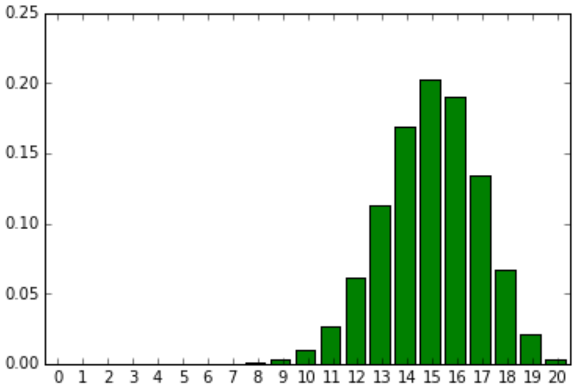
The Binomial Distribution

Binomial(n, p): # of heads from n independent coin tosses of bias (heads prob) p.



For $X \sim$ binomial(n, p),

$$\mathbb{E}X =$$

$$\mathrm{var}(X) =$$

$$\Pr(X = k) =$$

Fitting a Binomial Distribution to Data

Example: Survey on food tastes.

- You choose 1000 people at random and ask them whether they like sushi.
- 600 say yes.

What is a good estimate for the fraction of people who like sushi? Clearly, 60%.

More generally, say you observe n tosses of a coin of unknown bias, and k come up heads. What distribution binomial(n, p) is the best fit to this data?

Maximum Likelihood: a Small Caveat

You have two coins of unknown bias.

- You toss the first coin 10 times, and it comes out heads every time.
  You estimate its bias as p_1 =
- You toss the second coin 10 times, and it comes out heads once.
  You estimate its bias as p_2 =

Now you are told that one of the coins was tossed 20 times and 19 of them came out heads. Which coin do you think it is?

- Likelihood under p_1: $\Pr(19 \text{ heads out of } 20 \text{ tosses} | \text{bias} = 1) =$
- Likelihood under p_2: $\Pr(19 \text{ heads out of } 20 \text{ tosses} | \text{bias} = 0.1) =$

Laplace Smoothing

A smoothed version of maximum-likelihood: when you toss a coin n times and observe k heads, estimate the bias as

$$p = \frac{k+1}{n+2}.$$

We will later justify this in a Bayesian setting.

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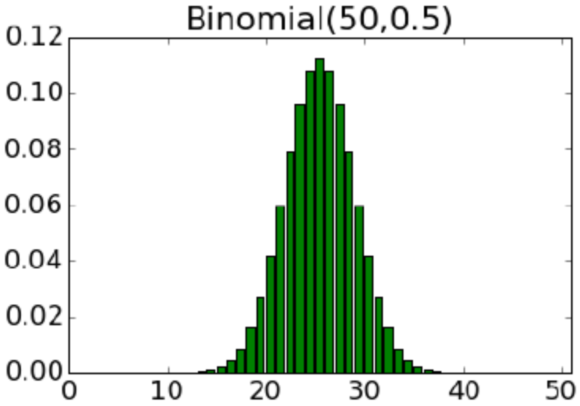
Laplace's law of succession: What is the probability that the sun won't rise tomorrow?

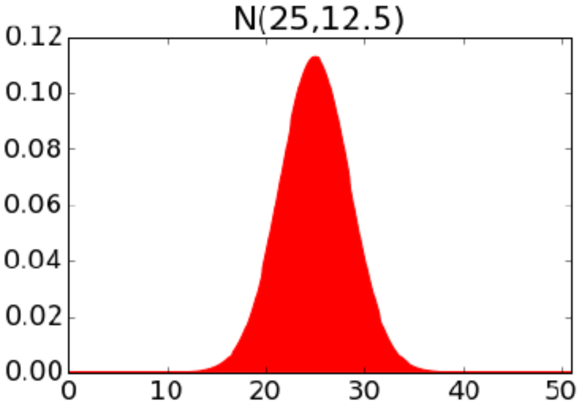
- Let p be the probability that the sun won't rise on a randomly chosen day. We want to estimate p.
- For the past 5000 years ($= \ 1825000$ days), the sun has risen every day. Using Laplace smoothing, estimate

$$p = \frac{1}{1825002}.$$

Normal Approximation to the Binomial



$\Longrightarrow$



When a coin of bias p is tossed n times, let S_n be the number of heads.

- We know S_n has mean np and variance $np(1 - p)$.
- By central limit theorem: As n grows, the distribution of S_n looks increasingly like a Gaussian with this mean and variance, i.e.,

$$\frac{S_n - np}{\sqrt{np(1-p)}} \xrightarrow{d} N(0,1).$$